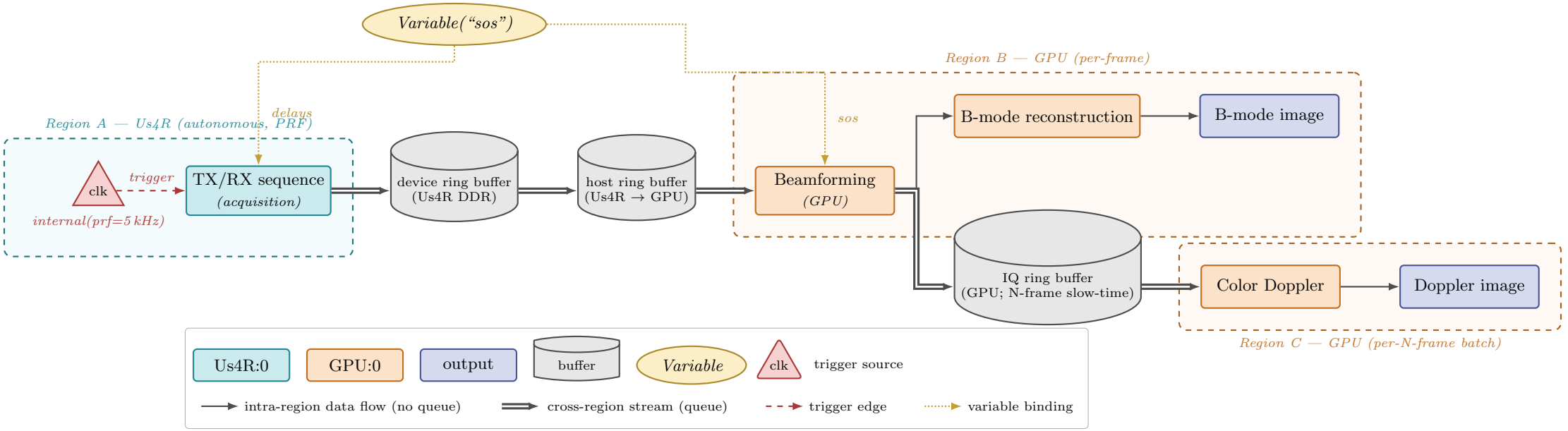

Variable("sos")
delays
Region A — Us4R (autonomous, PRF)
clk
trigger
internal(prf=5 kHz)
TX/RX sequence
(acquisition)
device ring buffer
(Us4R DDR)
host ring buffer
(Us4R → GPU)
Region B — GPU (per-frame)
sos
Beamforming
(GPU)
B-mode reconstruction
B-mode image
IQ ring buffer
(GPU; N-frame slow-time)
Color Doppler
Doppler image
Region C — GPU (per-N-frame batch)
Us4R:0
GPU:0
output
buffer
Variable
clk
trigger source
intra-region data flow (no queue)
cross-region stream (queue)
trigger edge
variable binding